

STM32 BASED PCB DESIGN

Pinanshu Ganesh Surve

Table of Contents

1. Introduction
2. STM32F103C8T6
3. Stm32 Cube IDE
 - 3.1. Pinout & Configuration
 - 3.2. Clock Configuration
4. Overview of the Design
5. Annotation
6. Footprint Assignment
7. PCB Layout & Routing
8. Conclusion

List of Figures

1. STM32F103CBTx Microcontroller Pinout Diagram
2. STM32CubeIDE Pinout & Configuration View
3. Schematic Diagram of the STM32-Based PCB
4. PCB Layout and Routing (Top Layer)
5. 3D Rendered View of the Assembled STM32-Based PCB

1. INTRODUCTION

This report details the comprehensive design process of an STM32-based Printed Circuit Board (PCB). The project encompasses all stages from initial schematic capture and component selection to the final PCB layout and preparation for manufacturing. The primary objective was to develop a compact and functional development board centered around the powerful STM32F103C8T6 microcontroller, suitable for a variety of embedded system applications. This document outlines the methodologies, tools, and design considerations employed to achieve a robust and efficient PCB design, demonstrating a practical application of electronics design principles.

2. STM32F103C8T6

The STM32F103C8T6 microcontroller is at the heart of this PCB design. This specific microcontroller, which is a member of STMicroelectronics' STM32F1 series, was chosen due to its excellent performance, peripheral set, and affordability, making it a great option for a variety of embedded applications. It has a powerful ARM Cortex-M3 32-bit RISC core that can run at up to 72 MHz, giving it enough processing power for a range of uses. Multiple timers, analog-to-digital converters (ADCs), digital-to-analog converters (DACs), and a variety of communication interfaces like SPI, I2C, and UART are examples of important integrated peripherals. Its selection was also influenced by its small LQFP-48 package and ample GPIOs (General Purpose Input/Output pins), which enable a small PCB design while offering a wide range of connectivity options. The 48 pins of the U1, STM32F103CBTx, including dedicated power (VDD, VSS, VDDA, VSSA), reset (NRST), boot mode (BOOT0), and multiple GPIOs designated PAX, PBx, and PCx—all necessary for connecting to external components and peripherals—are clearly shown in the pinout diagram.

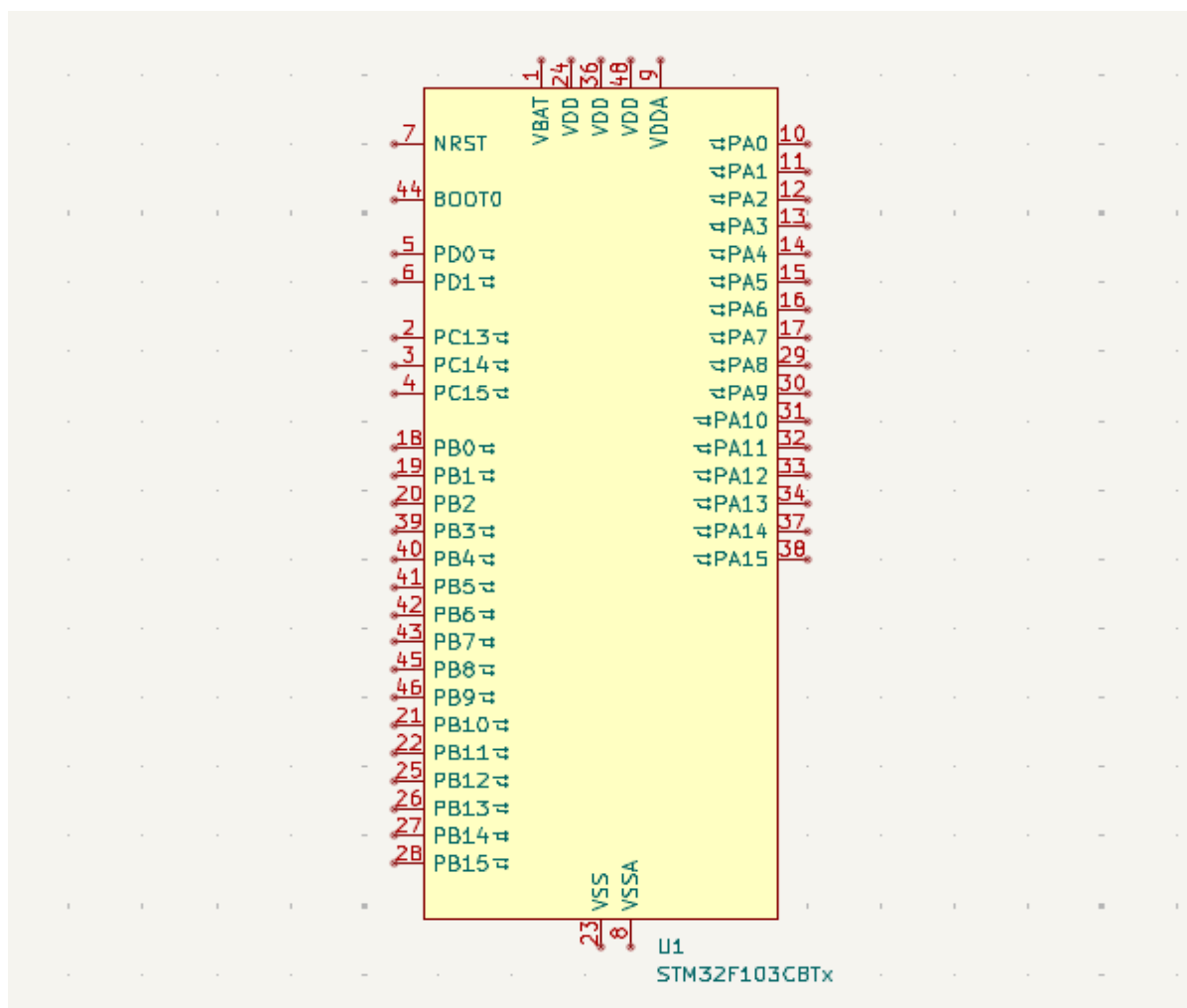


Figure 1 STM32F103CBTx Microcontroller Pinout Diagram

3. STM32 CUBE IDE

This project's design and development process made extensive use of STM32CubeIDE. Configuring the STM32F103C8T6 microcontroller, creating initialization code, and streamlining the software development process were all made possible by STMicroelectronics' extensive development environment. The project's initial setup phase was greatly streamlined by the STM32CubeIDE's user-friendly graphical interface for configuring clock configurations, pin assignments, and peripheral functionalities. Effective microcontroller coding, debugging, and programming were made possible by its integrated development tools, which included a C/C++ compiler, debugger, and project management capabilities. Verifying the functionality of the designed PCB was also made possible by this IDE's seamless integration with hardware debugging tools.

3.1.PINOUT AND CONFIGURATION

The exact pinout and configuration of the STM32F103C8T6 microcontroller, which is frequently done within STM32CubeIDE, was a crucial step in the PCB design process. Each pin of the U1 (STM32F103CBTx), as shown in the schematic, was meticulously matched to its particular function in accordance with the project specifications. While some PA pins (PA11, PA12) are devoted to USB connectivity, others (PA9, PA10) are routed for UART communication. The schematic clearly illustrates connections for a number of functions: BOOT0 is connected to a boot selection switch, and the NRST (reset) pin is connected to a reset switch (SW1) and related parts. Power supply pins that are connected to the proper power rails and have clear labels include VDD, VSS, VDDA, and VSSA. Additionally, the schematic highlights the intended communication and programming interfaces for the board by displaying the pins designated for I2C (PB6, PB7) and SWD (PA13, PA14) debugging. This careful pin mapping guarantees that the microcontroller can properly interface with all required external components and peripherals.

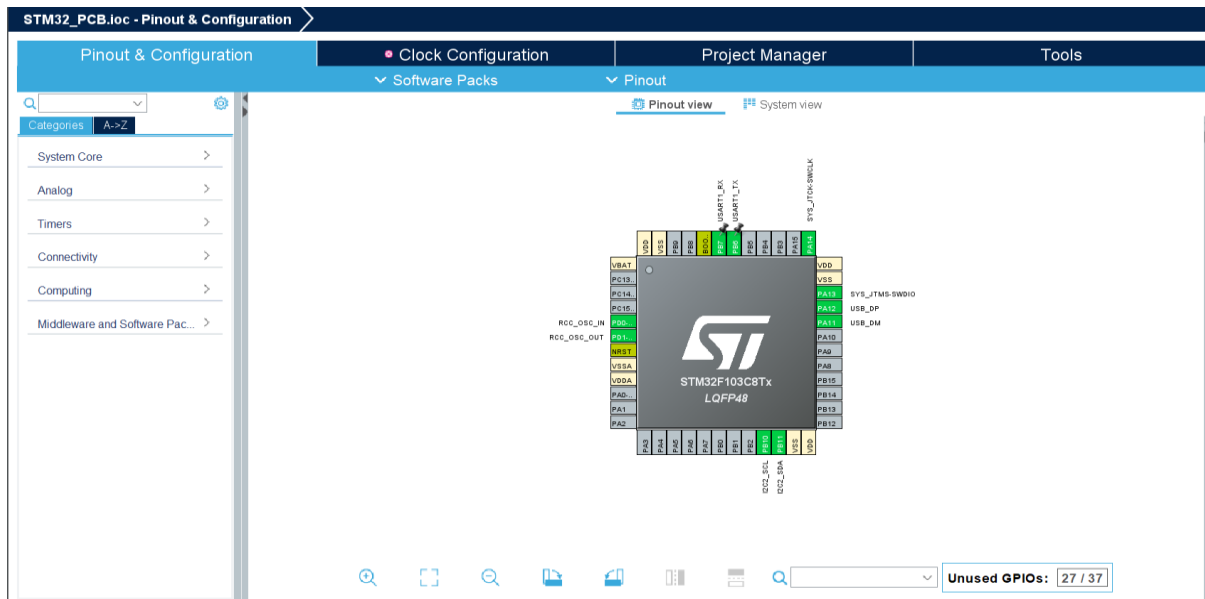


Figure 2 STM32CubeIDE Pinout & Configuration View

3.2.CLOCK CONFIGURATION

Accurate clock configuration is critical to the stability and functionality of any microcontroller-based system. The clocking scheme for the STM32F103C8T6 was meticulously created to guarantee peak performance. According to the schematic, the microcontroller's HSE_IN and HSE_OUT pins (PA0 and PA1, respectively) are connected to an external high-speed crystal oscillator (Y1, 8 MHz). The system receives a precise and reliable clock source from this external crystal. In order to generate the higher system clock frequency (up to 72 MHz) and derived clock frequencies for different peripherals, the design also takes into account the use of internal oscillators and the Phase-Locked Loop (PLL). The clock configuration maintains synchronization and dependable data transfer by ensuring that timers, the CPU, and communication interfaces all run at their designated speeds. A crucial component of the clock circuit, the 10 pF capacitors (C12, C13) are connected to the crystal for appropriate oscillation.

4. OVERVIEW OF THE DESIGN

The main schematic shows how this STM32-based PCB's overall design is organized into multiple separate functional blocks. These consist of the sections for the power supply, microcontroller and USB, and mounting holes. A voltage regulator (U2, AMS1117-3.3) is used in the "Power Supply" section to transform an input VBUS voltage (from a USB connector, nominal +5V) into a steady +3.3V supply for the entire board, which includes input (C1) and output (C2) filtering capacitors as well as an indicator LED (D1). The STM32F103C8T6 (U1) and its necessary decoupling capacitors (C3-C10), the external crystal oscillator (Y1) for clocking, and a reset switch (SW1) are located in the central "Microcontroller (and USB)" section. Additionally, the USB interface (J1, USB_Micro), related pull-up resistors (R3, R4), and filtering capacitors (C6, C7) are integrated for USB communication in this section. Additionally, the design includes specific connectors for SWD (J3), I2C (J4), and UART (J2) communication/debugging, which makes the board adaptable to a range of applications and development requirements. Additionally, mounting holes (H1-H4, M2) are provided for enclosure physical integration.

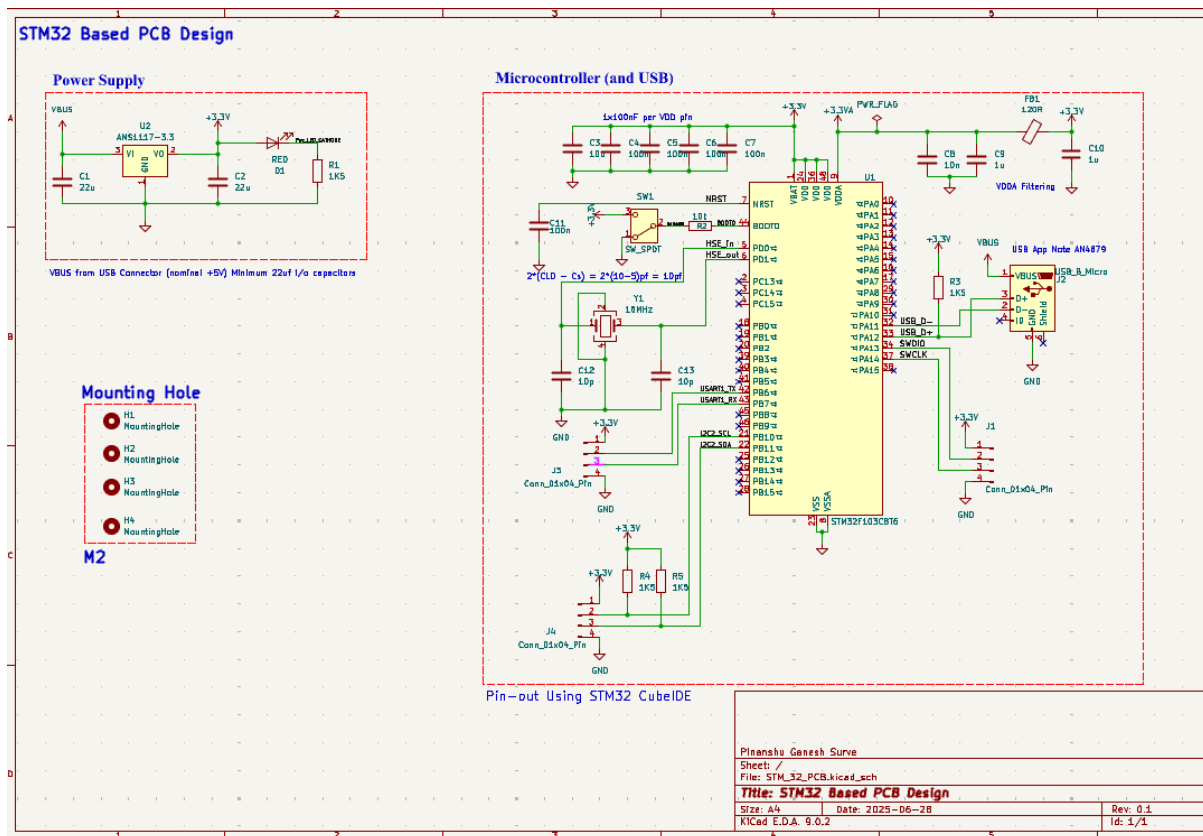


Figure 3 Schematic Diagram of the STM32-Based PCB

5. ANNOTATION

During the schematic design stage, annotation was an essential organizational step. This procedure entailed methodically giving each component in the schematic a unique reference designation (for example, U1 for the microcontroller, R1 for a resistor, and C1 for a capacitor). Each component is prominently labeled with its type and a sequential number, such as C1, C2, R1, R2, U1, U2, D1, and Y1, as can be observed in the schematics that are provided. Each component of the circuit is guaranteed to be easily identifiable, traceable, and clear thanks to this systematic labeling system. Annotating the schematic correctly reduces the possibility of errors during the PCB layout and assembly stages and is essential for debugging, troubleshooting, and facilitating communication between designers and manufacturers.

6. FOOTPRINT ASSIGNMENT

Footprint assignment was a crucial step after schematic capture and annotation. This entailed assigning a physical PCB footprint to each of the schematic's logical component symbols. For example, the voltage regulator (U2) was given its SOT-223 footprint, whereas the STM32F103C8T6 (U1) was given its LQFP-48 footprint. Based on their physical dimensions and power ratings, resistors and capacitors were given the proper surface-mount device (SMD) footprints, such as 0603 or 0805. Specific headers or connector footprints were assigned to connectors such as the USB Micro-B (J1), UART (J2), SWD (J3), and I2C (J4). To guarantee that the produced PCB can appropriately accommodate all components and that their pins line up with the copper pads, avoiding assembly problems and guaranteeing electrical connectivity, accurate footprint assignment is essential.

7. PCB LAYOUT AND ROUTING

The PCB layout and routing phase, which turned the logical schematic into a physical board, was the last and most visually appealing step of the design process. Components were positioned carefully to maximize signal integrity, reduce trace lengths, and guarantee effective power distribution, as shown in the PCB layout pictures. Essential supporting components, such as decoupling capacitors (C3-C10, C12, and C13) positioned near its power pins to filter noise, encircle the central STM32F103C8T6 (U1). The power section's U2 (voltage regulator) and related capacitors (C1, C2) are positioned to effectively supply the board with 3.3V. Important interface components are well-defined: the UART, SWD, and I2C headers are visible, as is the USB connector.

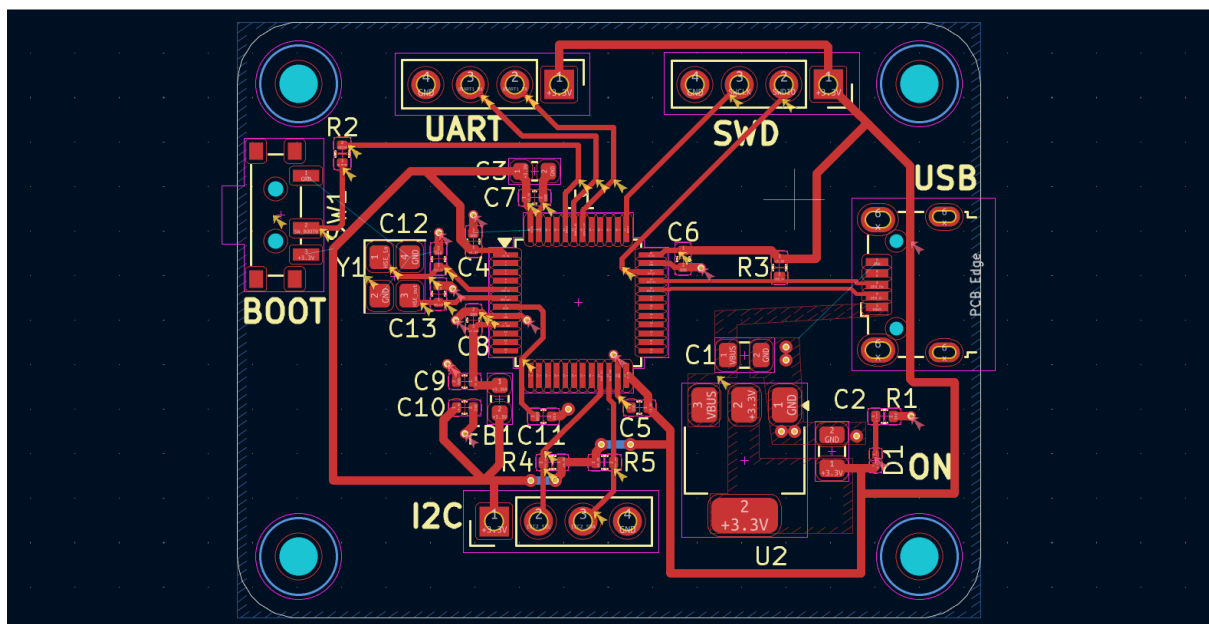


Figure 4 PCB Layout and Routing (Top Layer)

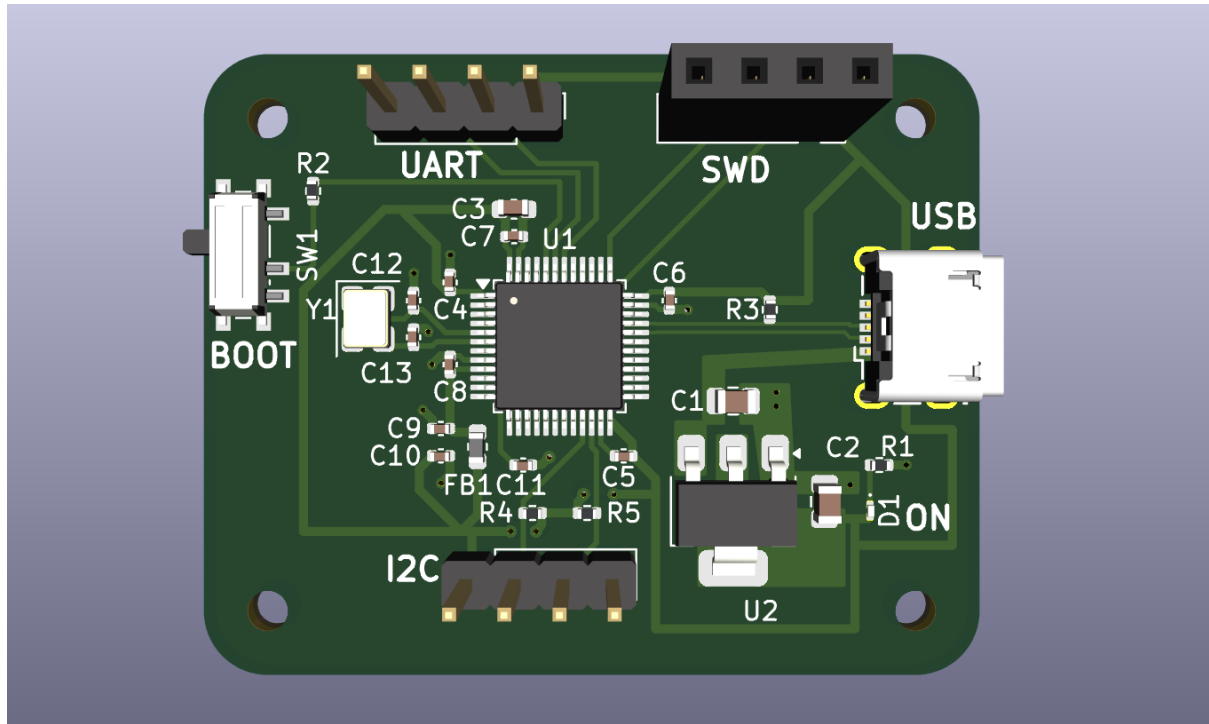


Figure 5 3D Rendered View of the Assembled STM32-Based PCB

Additionally, the BOOT switch (SW1) is positioned carefully. In order to connect each component pin in accordance with the netlist produced from the schematic, copper traces were painstakingly drawn on the board's layers (visible as red traces for the top layer). For stable operation, care was taken to maintain appropriate clearances, reduce crosstalk, and make sure that the ground and power planes (although not explicitly depicted as planes, the routing for VDD and GND is visible) were strong. The assembled board can be seen realistically in the final 3D render, which highlights the routing work and the neatly arranged component placement.

8. CONCLUSION

The entire process of creating an STM32-based PCB, from conceptualization and schematic design to physical layout and manufacturing readiness, was effectively illustrated by this project. With its strong-ARM Cortex-M3 core and adaptable peripheral set for a variety of embedded applications, the STM32F103C8T6 microcontroller was selected to provide a strong basis.

Efficient pinout configuration, accurate clock setup, and simplified software development were made possible by the methodical approach, which mainly relied on the STM32CubeIDE's capabilities. In order to guarantee the board's functionality, signal integrity, and manufacturability, crucial design factors were crucial, including the careful assignment of footprints and the thoughtful positioning and routing of components.

With its power supply, microcontroller, and multiple communication interfaces (UART, SWD, USB, and I2C), the finished PCB design offers a useful and adaptable foundation for the creation of future embedded systems. In addition to producing a physical hardware design, this project gave the participants invaluable experience with professional PCB design methodologies, from abstract schematic to a Gerber file set that is ready for production.